Heart Failure — Acute Decompensated HF (Table 265-1 Vasoactive Therapy)

TABLE 265-1 Vasoactive Therapy in Acute Decompensated Heart Failure

DRUG CLASS	GENERIC DRUG	USUAL DOSING	SPECIAL CAUTION	COMMENTS
Inotropic therapy				Use in hypotension, end-organ hypoperfusion, or shock states
	Dobutamine	2–20 μg/kg per min	Increased myocardial oxygen demand, arrhythmia	Short acting, an advantage; variable efficacy in presence of beta blockers (requires higher doses); clinical tolerance to prolonged infusions; concerns with hypersensitivity carditis (rare)
	Milrinone	0.375–0.75 μg/kg per min	Hypotension, arrhythmia	Decrease dose in renal insufficiency; avoid initial bolus; effectiveness retained in presence of beta blockers
	Levosimendan	0.1 μg/kg per min; range, 0.05–0.2 μg/kg per min	Hypotension, arrhythmia	Long acting; should not be used in presence of low blood pressure; similar effectiveness as dobutamine but effectiveness retained in presence of beta blockers
Vasodilators				Use in presence of pulmonary congestion for rapid relief of dyspnea, in presence of a preserved blood pressure
	Nitroglycerin	10–20 μg/min, increase up to 200 μg/min	Headache, flushing, tolerance	Most common vasodilator but often underdosed; effective in higher doses
	Nesiritide	Bolus 2 μg/kg and infusion at 0.01 μg/kg per min	Hypotension	Decrease in blood pressure may reduce renal perfusion pressure; bolus may be avoided since it increases hypotension predilection
	Nitroprusside	0.3 μg/kg per min titrated to 5 μg/kg per min	Thiocyanate toxicity in renal insufficiency (>72 h)	Requires arterial line placement for titration for precise blood pressure management and prevention of hypotension
	Serelaxin	N/A (tested at 30 μg/kg per d)	Baseline blood pressure should be >125 mmHg	Not widely commercially available; ineffective in confirmatory trials
	Ularitide	15 ng/kg per min (48 h)	Baseline blood pressure >116 mmHg	Excess hypotension and increased serum creatinine
Diuretics				First line of therapy in volume overload with congestion; may use bolus or continuous dosing; initial low dose (1 × home dose) or high dose (2.5 × home dose) equally effective with higher risk of renal worsening with higher dose
	Furosemide	20–240 mg daily	Monitor for electrolyte loss	In severe congestion, use intravenously and consider continuous infusion (not trial supported)
	Torsemide	10–100 mg daily	Monitor for electrolyte loss	High bioavailability, can be given orally; anecdotally more effective in advanced heart failure states if furosemide less bioavailable (due to gut congestion)
	Bumetanide	0.5–5 mg daily	Monitor for electrolyte loss	Can be used orally; intermediate bioavailability
	Adjuvant diuretics for augmentation	N/A	Metolazone, chlorthalidone, spironolactone, acetazolamide	Acetazolamide is useful in presence of alkalosis (bicarbonate level >27mEq/L); metolazone given in 2.5- to 10-mg doses; concomitant use of loop diuretics and thiazides associated with risk for severe hypokalemia, careful laboratory monitoring advised; spironolactone is useful in presence of severe hypokalemia and normal renal function

1. Inotropes — when to use

WHAT YOU SEE

The 'Inotropic therapy' class header — the top blue row grouping dobutamine, milrinone and levosimendan; the COMMENTS cell flags 'use in hypotension, end-organ hypoperfusion, or shock states', marking this whole block as the cold/shock-only tier.

WHAT IT ENCODES

Inotropes (dobutamine, milrinone, levosimendan) augment contractility and are reserved for the COLD/hypoperfused patient — hypotension with end-organ hypoperfusion (rising creatinine, cool extremities, altered mentation, lactic acidosis) or frank cardiogenic shock. They are a bridge to recovery, decision, device, or transplant — not a routine ADHF drug. They raise myocardial oxygen demand and are proarrhythmic, and observational data link routine inotrope use to higher mortality, so use the lowest effective dose for the shortest time.

BOARD TRAP

WRONG: inotropes are first-line therapy for all acute decompensated HF. Discriminator — most ADHF is WET-WARM (congested but well-perfused) and is treated with diuretics ± vasodilators; inotropes are reserved for the WET-COLD/low-output, hypotensive patient because they increase O2 demand, are arrhythmogenic, and are associated with increased mortality when used indiscriminately.

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2. Dobutamine

WHAT YOU SEE

Dobutamine row, 2-20 µg/kg/min

WHAT IT ENCODES

Dobutamine is a beta-1 (and weak beta-2) agonist given 2–20 µg/kg/min; it raises cardiac output and modestly vasodilates. Advantages: short half-life (~2 min), easily titrated. Drawbacks: increases myocardial O2 demand and is arrhythmogenic (sinus tachycardia, AF, ventricular ectopy); rare hypersensitivity myocarditis with prolonged use. Crucially, its effect is BLUNTED in patients on beta-blockers (it competes at the beta-receptor) — so higher doses may be needed in a beta-blocked patient.

BOARD TRAP

WRONG: dobutamine is the inotrope of choice for a patient on chronic high-dose beta-blocker therapy. Discriminator — beta-blockade competitively BLUNTS dobutamine (a beta-agonist); MILRINONE (a PDE3 inhibitor acting downstream of the beta-receptor) retains its effect in beta-blocked patients and is preferred there.

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3. Milrinone

WHAT YOU SEE

Milrinone row, 0.375-0.75 µg/kg/min

WHAT IT ENCODES

Milrinone is a phosphodiesterase-3 inhibitor (0.375–0.75 µg/kg/min) — an 'inodilator': it increases contractility AND vasodilates (pulmonary + systemic), lowering filling pressures and afterload. Because it acts DOWNSTREAM of the beta-receptor (raising intracellular cAMP), its effect is RETAINED in beta-blocked patients — its key advantage over dobutamine. Caveats: it is RENALLY cleared (accumulates and causes prolonged hypotension/arrhythmia in renal insufficiency — reduce dose) and causes more hypotension than dobutamine.

BOARD TRAP

WRONG: beta-blockers blunt milrinone's inotropic effect just like dobutamine. Discriminator — milrinone (PDE3 inhibitor) works DOWNSTREAM of the beta-adrenergic receptor, so its effect is PRESERVED despite beta-blockade — that is exactly why it is favored over dobutamine in beta-blocked patients.

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4. Levosimendan

WHAT YOU SEE

The levosimendan row — dosing cell reads 0.1 µg/kg/min (range 0.05–0.2) and the COMMENTS cell stresses 'long acting' and 'effectiveness retained in presence of beta blockers', the two features that define this calcium-sensitizer inodilator.

WHAT IT ENCODES

Levosimendan is a calcium SENSITIZER (0.1 µg/kg/min; range 0.05–0.2) — it increases the myofilament response to calcium (positive inotropy without raising intracellular calcium or O2 demand much) and opens K-ATP channels (vasodilation). It is long-acting (active metabolite gives days of effect) and, like milrinone, RETAINS efficacy with beta-blockers. It causes hypotension — avoid/use cautiously when BP is low. (Not FDA-approved in the US; used in Europe.)

BOARD TRAP

WRONG: levosimendan is safe to start in a hypotensive cardiogenic-shock patient because it 'sensitizes' rather than stimulates. Discriminator — levosimendan VASODILATES and worsens HYPOTENSION; avoid it when blood pressure is low. Its long half-life also means adverse effects (hypotension, arrhythmia) persist for days after stopping.

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5. Vasodilators — congestion + preserved BP

WHAT YOU SEE

The 'Vasodilators' class header row grouping nitroglycerin, nesiritide and nitroprusside — used for pulmonary congestion with preserved blood pressure (rapid relief of dyspnea).

WHAT IT ENCODES

Vasodilators (nitroglycerin, nitroprusside, nesiritide) reduce preload and afterload to rapidly relieve pulmonary congestion and dyspnea — best for the patient with congestion AND PRESERVED/elevated blood pressure (the hypertensive 'flash' pulmonary edema phenotype). They lower filling pressures fast, complementing diuretics. Absolute prerequisite: adequate blood pressure — they are contraindicated in hypotension and in severe aortic stenosis.

BOARD TRAP

WRONG: vasodilators are safe in the hypotensive ADHF patient. Discriminator — vasodilators require PRESERVED blood pressure; giving nitroglycerin/nitroprusside to a hypotensive (cold/low-output) patient precipitates dangerous hypotension and worsens organ perfusion. They shine in HYPERTENSIVE acute pulmonary edema, not in cardiogenic shock.

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6. Nitroglycerin

WHAT YOU SEE

Nitroglycerin row, 10-20 µg/min

WHAT IT ENCODES

IV nitroglycerin (start 10–20 µg/min, titrate up to ~200 µg/min) is predominantly a VENODILATOR at low doses — it drops preload and pulmonary congestion quickly, with arterial (afterload) effects emerging at higher doses. First-line vasodilator in hypertensive acute pulmonary edema. Limitations: throbbing HEADACHE and flushing, and TACHYPHYLAXIS/tolerance within ~24–48 h of continuous infusion. Avoid within 24–48 h of a PDE5 inhibitor (sildenafil/tadalafil) — risk of profound hypotension.

BOARD TRAP

WRONG: nitroglycerin can be given to a patient who recently took sildenafil for erectile dysfunction. Discriminator — combining nitrates with a PDE5 inhibitor (sildenafil 24 h, tadalafil 48 h) causes life-threatening hypotension and is CONTRAINDICATED; always ask about recent PDE5-inhibitor use before starting nitroglycerin.

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7. Nesiritide

WHAT YOU SEE

The nesiritide row — COMMENTS cell bluntly states 'not widely commercially available; ineffective in confirmatory trials', the visual tag that this BNP analog is a failed drug.

WHAT IT ENCODES

Nesiritide is recombinant B-type natriuretic peptide (BNP) — a vasodilator that lowers filling pressures and promotes natriuresis. The initial bolus may be omitted to limit hypotension. Despite the appealing mechanism, the ASCEND-HF trial showed NO mortality/rehospitalization benefit and more hypotension; it is no longer commercially available/used. Board point: a mechanistically attractive drug that FAILED in trials.

BOARD TRAP

WRONG: nesiritide reduces mortality and rehospitalization in ADHF, so it should be added routinely. Discriminator — nesiritide (recombinant BNP) showed NO outcome benefit and MORE hypotension in trials (ASCEND-HF) and is no longer used; mechanism does not equal proven benefit.

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8. Nitroprusside — renal caution

WHAT YOU SEE

Nitroprusside row, thiocyanate

WHAT IT ENCODES

Sodium nitroprusside (0.3 µg/kg/min titrated up to ~5 µg/kg/min) is a potent balanced arterial AND venous dilator — ideal for severe hypertensive crisis with pulmonary edema or acute mitral/aortic regurgitation (afterload reduction). It requires an ARTERIAL LINE for minute-to-minute BP titration. It is metabolized to CYANIDE then THIOCYANATE; thiocyanate accumulates and causes toxicity (confusion, seizures, lactic acidosis) in RENAL insufficiency and with prolonged infusion (>24–72 h). Also causes coronary steal in ischemia.

BOARD TRAP

WRONG: nitroprusside is safe for prolonged use in a patient with renal failure. Discriminator — nitroprusside generates THIOCYANATE (and cyanide) that accumulate in RENAL insufficiency and with infusions beyond ~24–72 h, causing potentially fatal toxicity; limit duration, monitor levels, and avoid in significant renal impairment.

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9. Serelaxin / ularitide — ineffective

WHAT YOU SEE

The serelaxin and ularitide rows — both list a baseline-BP prerequisite (>125 / >116 mmHg) and the COMMENTS flag 'not widely commercially available; ineffective in confirmatory trials', marking them as failed investigational vasodilators.

WHAT IT ENCODES

Serelaxin (recombinant relaxin-2, given when BP >125 mmHg) and ularitide (synthetic urodilatin/natriuretic peptide, BP >116 mmHg) are investigational vasodilators that produced exciting early signals but were NEGATIVE in confirmatory phase-3 trials (RELAX-AHF-2, TRUE-AHF) — no improvement in cardiovascular death or worsening HF. Neither is commercially available. Another reminder that ADHF vasodilator outcome trials have repeatedly disappointed.

BOARD TRAP

WRONG: serelaxin and ularitide improve survival in ADHF based on their natriuretic/vasodilatory mechanism. Discriminator — both FAILED their confirmatory outcome trials and are not commercially available; positive early-phase signals did not translate into mortality benefit.

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10. Diuretics — first line for volume overload

WHAT YOU SEE

The 'Diuretics' class header row — the COMMENTS cell reads 'first line of therapy in volume overload with congestion', flagging loop diuretics as the wet-patient cornerstone.

WHAT IT ENCODES

Loop diuretics are FIRST-LINE for the congested/volume-overloaded patient (the wet phenotype) — they relieve dyspnea and edema by inhibiting the Na-K-2Cl transporter in the thick ascending limb. Dosing: IV (oral GI absorption is unreliable when the gut is congested), starting at ~1× the home oral dose up to ~2.5× home dose; the DOSE-AHF trial showed low- vs high-dose and bolus vs continuous infusion were broadly equivalent, with higher doses giving faster relief at the cost of transient creatinine bumps. Monitor electrolytes (K, Mg) and renal function.

BOARD TRAP

WRONG: in a congested ADHF patient you should start with oral diuretics and inotropes. Discriminator — give diuretics INTRAVENOUSLY in decompensated congestion (gut edema impairs oral absorption); inotropes are NOT for the well-perfused congested (wet-warm) patient — they're reserved for the cold/hypoperfused patient.

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11. Furosemide IV in severe congestion

WHAT YOU SEE

Furosemide row, IV in severe

WHAT IT ENCODES

Furosemide (20–240 mg/day) is the prototype loop diuretic; oral bioavailability is variable (~10–100%, averaging ~50%) and worsens with gut wall edema. In severe congestion give it IV; if response is inadequate, switch to a CONTINUOUS INFUSION (after a loading bolus) to maintain steady tubular drug levels. Monitor for hypokalemia, hypomagnesemia, metabolic alkalosis (contraction), ototoxicity at high IV doses, and worsening renal indices. IV-to-PO conversion for furosemide is roughly 1:2 (40 mg IV ≈ 80 mg PO).

BOARD TRAP

WRONG: oral furosemide gives the same predictable diuresis as IV furosemide in a congested patient. Discriminator — oral furosemide absorption is erratic and BLUNTED by gut edema in ADHF; use IV dosing in significant congestion and recall the ~1:2 IV-to-oral potency conversion when transitioning home.

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12. Torsemide better absorbed

WHAT YOU SEE

Torsemide row, advanced HF

WHAT IT ENCODES

Torsemide (10–100 mg/day) has MORE reliable, near-complete oral bioavailability (~80–100%) and a longer half-life than furosemide, so its oral effect is consistent even when gut congestion limits furosemide absorption — anecdotally useful in advanced HF and frequent readmitters. (The TRANSFORM-HF trial found no mortality difference vs furosemide, but bioavailability/dosing convenience still drives the switch in selected patients.) Torsemide ~20 mg ≈ furosemide ~40 mg PO ≈ bumetanide ~1 mg.

BOARD TRAP

WRONG: oral furosemide and torsemide have equal bioavailability, so they are interchangeable milligram-for-milligram. Discriminator — torsemide (and bumetanide) have much more reliable oral absorption than furosemide and are roughly twice as potent (torsemide 20 mg ≈ furosemide 40 mg); failing to dose-convert when switching causes under- or over-diuresis.

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13. Adjuvant — metolazone/acetazolamide

WHAT YOU SEE

Adjuvant diuretic augmentation row

WHAT IT ENCODES

For diuretic resistance, add a SECOND nephron segment ('sequential nephron blockade'): a thiazide-type agent — METOLAZONE (2.5–10 mg, retains effect in renal impairment) or chlorthalidone — blocks distal Na reabsorption and powerfully augments loop diuresis; ACETAZOLAMIDE is especially useful when there is metabolic ALKALOSIS (serum bicarbonate >27 mEq/L), as shown in the ADVOR trial where adding acetazolamide improved decongestion. Spironolactone helps with hypokalemia/normal renal function. Watch for marked HYPOKALEMIA and volume depletion with combination diuresis.

BOARD TRAP

WRONG: when diuresis stalls you should just keep escalating the loop diuretic alone. Discriminator — combination/sequential nephron blockade (add metolazone/thiazide, or acetazolamide if bicarbonate >27/alkalotic) overcomes loop-diuretic resistance far better than ever-higher loop doses, but mandates close potassium and volume monitoring to avoid severe hypokalemia.